



Indian Association for the Cultivation of Science

(Deemed to be University under the *de novo* category)

Integrated Bachelor's – Master's program

Mid-term Examination - Autumn 2025

Subject: Introduction to Computing

Subject Code(s): COM 1101

Full marks: 25

Time allotted: 2 hrs

Q1. Explain the output of the programs. If there is any mistake in any program that may lead to compilation error, mention the necessary correction(s) that will lead to error-free compilation. {5 × 3 Marks}

- i) Refer to the program written in Figure 1. Rewrite this code, to reflect the modifications after the pre-processing step and subsequently explain the output of the program.

```
1 #include<stdio.h>
2 #define MIN(a,b) (a<b)?a:b
3
4 int main(void)
5 {
6     int m = 10*MIN(2+4,5);
7     printf("Min is %d\n", m);
8     return 0;
9 }
```

Figure 1

```
1 #include<stdio.h>
2
3 int main(void)
4 {
5     int m = 12345678901;
6     printf("%u",printf("%d\n", m));
7     return 0;
8 }
```

Figure 2

- ii) Refer to the program written in Figure 2. Explain the output.
- iii) Refer to the program written in Figure 3. Explain the output.
- iv) Refer to the program written in Figure 4. Explain the output. Explain if the `break` provided in the default case is mandatory in this program.

```

1 #include<stdio.h>
2
3 int main(void)
4 {
5     int i = 3;
6     do{
7         printf("iter %d\n",i);
8         i--;
9     }while(i = 0);
10    printf("loop finished\n",i);
11    return 0;
12 }

```

Figure 3

```

1 #include<stdio.h>
2
3 int main(void)
4 {
5     int i = 1, option = 2;
6     switch(option){
7         case 1:
8         case 2: if(!i) break;
9         case 3:
10        case 4: printf("%d", ++i);
11                break;
12        default: break;
13    }
14    return 0;
15 }

```

Figure 4

v) Refer to the program written in Figure 5. Explain the output.

```

1 #include<stdio.h>
2
3 int main(void)
4 {
5     int i=-1, j=-1, k=0, l=2, m;
6     m = i++ && k++ && j++ || l++;
7     printf("%d %d %d %d %d",i,j,k,l,m);
8     return 0;
9 }

```

Figure 5

Q2. Represent 23.390625 into IEEE-754 32-bit (single-precision) format. Explain the offset value for such a system and its importance. {4 Marks}

Q3. Write a C program to output the sum of the following series up to  $n$  terms. The program should be able to scale up for a large  $n$ .  $S = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$  {4 Marks}

Q4. What is `sizeof()`? What will be the output of the following lines in C,  
`int a[5];`

`printf("%d",sizeof(a));`

{1+1 = 2 Marks}